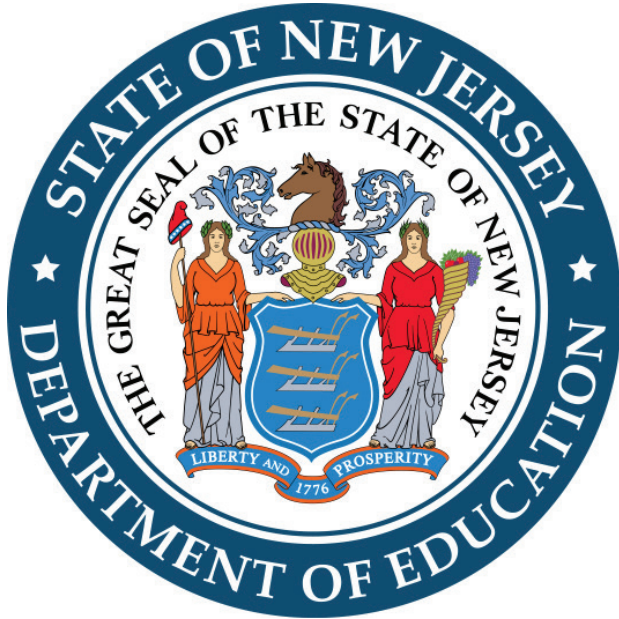




New Jersey
Student Learning
Assessments-Adaptive

Practice Test

Algebra II

Student Name _____



Directions:

Today, you will take the Algebra II Practice Test. You will be able to use a calculator. If you do not know the answer to a question, you may go on to the next question. When you finish, you may review your answers and any questions you did not answer.

1. For questions with bubbled responses, fill in the circle next to your answer choice. If you change your answer, make sure you erase your old answer completely. Do not cross out or make any marks on the other choices.
2. For questions with response boxes, write your answer neatly, clearly, and in the space provided.
3. For questions with response grids, write your answer in the answer boxes at the top of the grid.
 - Write only one digit or symbol in each answer box.
 - Use only the options listed below each box.
 - Be sure to write a negative sign or decimal point in the answer box if it is part of the answer.

4	3							} Answer boxes } Negative sign } Decimal point } Number bubbles
-	-	-	-	-	-	-	-	
.	.	.	.	.	.	.	.	
0	0	0	0	0	0	0	0	
1	1	1	1	1	1	1	1	}
2	2	2	2	2	2	2	2	
3	●	3	3	3	3	3	3	
●	4	4	4	4	4	4	4	
5	5	5	5	5	5	5	5	
6	6	6	6	6	6	6	6	
7	7	7	7	7	7	7	7	
8	8	8	8	8	8	8	8	
9	9	9	9	9	9	9	9	



1. A function with a factor of $x - 4$ is shown.

$$f(x) = x^4 - 3x^3 - 7x^2 + ax + 8$$

What is the value of a ?

-	-	-	-	-	-	-	-
.	.	.	.	.	.	.	.
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

2. A logarithmic equation is shown.

$$\log_2 17 = n - 5$$

Create an equivalent equation in exponential form.



3. Both $f(x)$ and $g(x)$ are polynomial functions.

Determine whether each expression must be a polynomial.

	Must Be a Polynomial	May Not Be a Polynomial
$f(x) + g(x)$	(A)	(B)
$g(x)[f(x) - g(x)]$	(C)	(D)
$\frac{f(x) - g(x)}{f(x)}$	(E)	(F)

4. The results of a country's transportation study for the year 2010 are shown.

- The average fuel efficiency of all the vehicles, which is the average miles traveled per gallon of gasoline (MPG), was m miles per gallon.
- The total miles traveled can be estimated as 135 billion miles.
- The average cost of gasoline in 2010 was \$2.84 per gallon.

Create a function $S(m)$ to show how much money, in billions of dollars, would have been saved if the MPG was $m + 1$ miles per gallon in the year 2010. Assume the same values for total miles traveled and cost of gasoline.

$S(m) =$



5. A store manager records the number of customers at the store each week for several years. The data are approximately normally distributed with a mean of 1245 and a standard deviation of 35.

The manager wants to predict future customer totals.

Using these data, what percentage of weeks will have between 1280 and 1315 customers?

							%
-	-	-	-	-	-	-	
.	.	.	.	.	.	.	
0	0	0	0	0	0	0	
1	1	1	1	1	1	1	
2	2	2	2	2	2	2	
3	3	3	3	3	3	3	
4	4	4	4	4	4	4	
5	5	5	5	5	5	5	
6	6	6	6	6	6	6	
7	7	7	7	7	7	7	
8	8	8	8	8	8	8	
9	9	9	9	9	9	9	

6. A function is shown.

$$f(x) = x^3 - 5x^2 - 4x + 20$$

What is the end behavior of the graph of $f(x)$ as x increases?

- (A) $f(x)$ approaches zero.
- (B) $f(x)$ approaches 20.
- (C) $f(x)$ approaches negative infinity.
- (D) $f(x)$ approaches positive infinity.



Mathematics

7. A quadratic equation is shown.

$$0 = 2m^2 + 3m + 7$$

What is one exact solution to the equation?

$m =$

8. An expression is shown.

$$3i^2 + \sqrt{-121} - \sqrt{49}$$

Create an equivalent expression in the form $a + bi$.



9. A function is shown.

$$g(c) = c(4c + 8)(c - 1)$$

Select all of the values of c that are zeros of the function.

- (A) -2
- (B) -1
- (C) 0
- (D) 1
- (E) 2
- (F) 4

10. Julian graphs the function $f(x) = 2^x + 5$. He then moves the graph down 8 units to create function $g(x)$.

Create an equation that represents $g(x)$.

$g(x) =$

11. Which expression is equivalent to $(2r^2 + r - 1) - (3r^2 + 4r - 5)$?

- (A) $-r^2 - 3r + 4$
- (B) $-r^2 + 5r - 6$
- (C) $5r^2 - 3r + 4$
- (D) $5r^2 + 5r - 6$



- 12.** A survey of 525 people was conducted to determine whether they have brothers and sisters.
- The results showed that 24% of the people surveyed do not have a sister and 68% of the people surveyed have a brother.
 - The results also showed that 93 of the people surveyed do not have a sister and do not have a brother.

Complete the two-way frequency table to show the results of the survey.

Survey Results

	Have a Brother	Do Not Have a Brother	Total
Have a Sister			
Do Not Have a Sister			
Total			525

- 13.** An expression is shown.

$$64x^2 - 196$$

Michael rewrites this expression in a different form.

Which form could Michael have used, where a and b are integers?

- Ⓐ $(ax - b)^2$
- Ⓑ $(ax + b)^2$
- Ⓒ $(ax + b)(ax - b)$
- Ⓓ $(ax + b)(bx - a)$



Paper Practice Test Answer and Alignment Document: Math

Algebra II

Item Sequence	Answer Key	Standards Alignment
1	10, or any equivalent value	A.APR.B.2
2	$2^{n-5} = 17$, or any equivalent equation	F.LE.A.4
3	Row 1: Must Be a Polynomial Row 2: Must Be a Polynomial Row 3: May Not Be a Polynomial	A.APR.A.1
4	$2.84(\frac{135}{m} - \frac{135}{m+1})$, or any equivalent function	F.BF.A.1b
5	13.5, or any equivalent value	S.ID.A.4
6	D	F.IF.C.7c
7	$\frac{-3+i\sqrt{47}}{4}$ or $\frac{-3-i\sqrt{47}}{4}$, or any equivalent expressions	N.CN.C.7
8	$-10 + 11i$, or any equivalent expression containing -10 and $11i$	N.CN.A.2
9	A, C, D	A.APR.B.3
10	$2^x - 3$, or any equivalent expression	F.BF.B.3
11	A	A.APR.A.1
12	Row 1: 324, 75, 399 Row 2: 33, 93, 126 Row 3: 357, 168 or any equivalent values	S.ID.B.5
13	C	A.SSE.A.2